GyaanConnect

MASTER PROJECT AUDIT & ROADMAP

"Don't just watch the lesson. Prove that you understood it."

Voice-First Diagnostic Learning Platform for Indian School Students (Classes 1–12 • CBSE, ICSE, UPMSP & State Boards)

AUDIT DATE	REPOSITORY	AUDIT SCOPE	COMPLIANCE STATUS
October 15, 2026	GYAAN (React 19 + Node.js)	Frontend, Backend, Security, AI, Data	Phase 1 Audit Complete

1. Current Architecture Overview

The platform is structured as a client-centric single-page application paired with a lightweight Node.js API service:

- **Frontend Layer:** React 19 (`^19.2.8`), Vite 8 (`^8.2.2`), Tailwind CSS v4 (`^4.3.3`), Lucide icons, and Canvas Confetti.
- **Backend Layer:** Native Node.js ESM HTTP service (`server/index.js`) on port 3001, reverse-proxied in development via Vite (`/api` → `http://localhost:3001`).
- **3-Tier AI Evaluation Pipeline:**
  • *Tier 1 (Server Proxy):* `POST /api/evaluate` calls Google Gemini API with server-side credentials.
  • *Tier 2 (Client Fallback):* Optional client key entered in modal (stored in localStorage; marked for removal).
  • *Tier 3 (Local Socratic Engine):* 100% offline, zero-cost heuristic evaluator matching must-know points and detecting misconceptions.
- **Pedagogical DAG Engine:** Directed Acyclic Graph dependency model (`conceptGraphEngine.js`) mapping ancestral prerequisites and diagnosing foundational learning gaps.
- **Spaced Repetition Scheduler:** Implementation of the SuperMemo SM-2 algorithm scheduling review sessions at 1-day, 3-day, and 7-day intervals based on student accuracy.
- **Native Audio Synthesis:** Web Audio API oscillators producing zero-latency auditory feedback for clicks, success chords, and error tones without external MP3 dependencies.

2. Working Features Matrix

Feature Module	Implementation Files	Status	Verified Capabilities
Feynman Oral Recall Studio	`src/components/ProveUnderstandingStudio.jsx`	WORKING	4-phase workflow (briefing → explaining → evaluating → evaluated); 5-pillar scores; confetti reward.
Voice Recognition & Synthesis	`src/services/aiVoiceService.js`	WORKING	Web Speech API for Indian English (`en-IN`) and Hindi (`hi-IN`); SpeechSynthesis tutor read-aloud; quiet text fallback.
Concept Graph Lineage	`src/components/ConceptGraphModal.jsx`	WORKING	DAG visualizer; identifies prerequisite health; warns when foundational concepts have <60% mastery.
Spaced Repetition Queue	`src/components/DashboardOverview.jsx`	WORKING	SM-2 algorithm calculates decay; dashboard displays overdue review items with 1-click oral retry.
Lecture & Study Theater	`src/components/LectureTheater.jsx`	WORKING	Curated video lectures, one-shot marathons, tab switching across Feynman studio, notes, quizzes, and concepts.
Practice MCQs & Gamification	`PracticeQuiz.jsx` , `gamificationService.js`	WORKING	Interactive MCQ arena, instant explanations, anti-spam XP awarding, daily streaks, level progression.
Local Data Backup Export/Import	`StudentProfileModal.jsx` , `studentService.js`	WORKING	One-click JSON file download and restore for local student progress and badges.
Global Search (Ctrl+K)	`src/components/GlobalSearchModal.jsx`	WORKING	Unified modal searching across concepts, chapters, subjects, and boards with keyboard shortcut.

3. Confirmed Errors & Missing Features

🚨 **Linter Failure: `npm run lint` Exits with Code 1**

Root Cause: `vite.config.js` defines `cacheDir: './.vite_cache'` , but `.oxlintrc.json` and `.gitignore` do not exclude `.vite_cache` . Oxlint scans pre-bundled vendor dependencies (`lucide-react.js`), failing with **3,186 warnings and 14 errors**.

Targeted command: `npx oxlint src server` succeeds with 0 errors and 2 minor effect warnings.

⚠️ **Missing Backend Routes: HTTP 404 on Curriculum API Calls**

`curriculumService.js` queries `GET /api/curriculum/subjects` and `GET /api/curriculum/chapters` , but neither endpoint exists in `server/index.js` . The server returns 404, which the client silently catches and falls back to local data.

> ⚠️ **Conflicting Board Definitions & Procedural Chapter Fabrication**
> **Data Fragmentation:** `conceptKnowledgeBase.js` defines 9 boards, `curriculumDatabase.js` defines 5 boards, and `curriculumData.js` defines only 3 boards. Selecting Maharashtra, Rajasthan, or Bihar causes `getGradeData()` to default back to UP Board.
> **Mock Masquerading:** Only Class 10 Science & Math have authentic syllabi. For all other grades (Classes 1–9, 11, 12), `generateChaptersForSubject()` synthesizes dummy titles ("*Chapter 1: Mathematics - Unit 1*") and generic quizzes.

> ℹ️ **Dead / Orphaned Code**
> `src/components/AiVoiceRoom.jsx` (584 lines, 24.8 KB) is completely unreferenced and superseded by `ProveUnderstandingStudio.jsx`.

Missing Features Against Master Project Requirements:

- **Materials & Practical Instructions:** No structured lab apparatus list, safety guidelines, or step-by-step experiment instructions.
- **Real-World Examples & News Links:** No connection of curriculum concepts to contemporary science news (e.g. ISRO missions).
- **Server-Side Progress Persistence:** No user database; all data is strictly in browser `localStorage`.
- **Automated Testing:** Zero unit, integration, or end-to-end tests exist in the project.

4. Security & Privacy Audit

> 🚨 **Critical: Insecure Evaluation Rubric Injection (Prompt Tampering)**
> In `server/index.js` (lines 38–60), the backend blindly constructs the evaluation prompt using `concept.mustKnowPoints` provided in the client's HTTP request body:
>
> ```
> Must-Know Core Points: ${JSON.stringify(targetConcept.mustKnowPoints || [])}
> ```
>
> A student or malicious script can send arbitrary `mustKnowPoints` matching their exact words to score 100%. The server **must** look up trusted rubrics from `curriculumDatabase.js` using `conceptId`.

> ⚠️ **Client-Side API Key Storage & Direct Gemini Calls**
> `GeminiSettingsModal.jsx` saves Gemini API keys to `localStorage`, and `aiVoiceService.js` directly initiates raw browser fetch requests to Google's API endpoint. This risks credential exposure to browser extensions and shared school PCs.

> ✅ **Child Privacy & Audio Retention Compliance**
> Audio is never recorded, retained, or transmitted to third-party cloud servers. Speech recognition occurs entirely in browser memory through the native Web Speech API.

5. Build and Test Verification Results

Verification Command	Execution Output	Status	Actionable Conclusion
`npm run lint`	Scanned 36 files; 3,186 warnings, 14 errors.	FAILED	Fails because `.vite_cache` is scanned. Must exclude cache from oxlint.
`npx oxlint src server`	27 files checked; 0 errors, 2 warnings.	PASSED	Core source code is clean; only 2 cascading `setState` warnings in effects.
`npm run build`	1,872 modules transformed; completed in 6.56s.	PASSED	Vite production bundle generated successfully (JS: 469.75 kB, CSS: 68.71 kB).
`node server/index.js`	Listens on port 3001; health returns HTTP 200.	PASSED	Server starts cleanly; returns HTTP 503 fallback when `GEMINI_API_KEY` is blank.
`npm test`	Script missing from `package.json`.	MISSING	Zero automated unit, integration, or E2E tests exist in the project.

6. Recommended Phased Implementation Roadmap

Phase 1: Stabilization & Security Hardening (Current Priority)

- Exclude `.vite_cache` and `dist` from `.oxlintrc.json` and `.gitignore` so `npm run lint` passes with 0 errors.
- Fix cascading `setState` calls in `ConceptSelectionGrid.jsx` and `ProveUnderstandingStudio.jsx`.
- Hardcode server rubric lookup: enforce that `POST /api/evaluate` retrieves authoritative concepts via `conceptId`.
- Implement missing `/api/curriculum/subjects` and `/api/curriculum/chapters` endpoints in `server/index.js`.
- Safely remove dead code (`AiVoiceRoom.jsx`).
- Install Vitest and setup test harness for scoring, DAG prerequisite traversal, and API response validation.

Phase 2: Curriculum Schema & Data Unification

- Merge competing Board definitions into a single authoritative schema.
- Eliminate fake procedural chapter loops; label uncurated topics as `draft` or `curation_pending`.
- Attach official syllabus provenance (NCERT Rationalised 2026-27 citations).

Phase 3: Voice Learning & Studio Accessibility (WCAG 2.1 AA)

- Add accessible ARIA live regions, clear microphone recording pulse indicators, and full keyboard navigation.
- Enhance quiet-room text-input mode and provide pause/stop/replay speech controls.

Phase 4: Socratic AI & Diagnostic Evaluation Engine

- Upgrade backend integration to current standard Gemini models with structured JSON schema validation.
- Multi-turn Socratic follow-up intuition probes; student feedback error-reporting button.

Phase 5: Student Progress & Persistent Cloud Synchronization

- Transition to optional server persistence with clear distinction between local and cloud progress.
- Authentic completion criteria (never mark complete solely on view).

Phase 6: Practical Lab Activities & Real-World Science News

- Integrate CBSE/UP Board Science practical apparatus, safety instructions, and step-by-step procedures.
- Curated age-appropriate science news links (ISRO, renewable technology, scientific discoveries).

Phase 7: Multi-Board Expansion & Official Verification

- Progressive verified rollout: CBSE → UP Board (Bilingual) → ICSE → Major State Boards.

7. The Smallest Safe Next Task

Zero-Risk Pipeline & Codebase Health Stabilization

1. Add `".vite_cache"` and `"dist"` to `.gitignore` and configure `.oxlintrc.json` ignore patterns.
2. Fix the two cascading `setState` warnings in `ConceptSelectionGrid.jsx` and `ProveUnderstandingStudio.jsx`.
3. Safely remove the unused, orphaned `src/components/AiVoiceRoom.jsx` file.
4. Run `npm run lint` and `npm run build` to confirm **100% green status with 0 errors and 0 warnings**.

Estimated Time: ~5 minutes. Modifies zero core application behavior; provides a rock-solid, error-free foundation.
